

A Phase-First Hamiltonian from Hilbert-Kahler Geometry, Bloch-Sphere Holonomy, Zeta-Polar Tetrahedral Depth, and Twin-Prime Collatz Dynamics

Formal working note for the Adrian-CIEL/NOEMA framework

2026-06-19

Abstract

This note rebuilds the phase-first Hamiltonian from the ground up. The starting point is a two-level Hilbert space whose projective space is the Kahler manifold $\mathbb{CP}^1$, equivalently the Bloch sphere with the Fubini–Study metric and Berry curvature. The standard geometric layer is then extended by model-specific axioms: an Euler/spin constraint selecting a spin-1/2 double-valued sector, zeta-zero labels at the spin poles, a tetrahedral depth frame inside the Bloch sphere, a Poincare-disk interference surface, and a Collatz rhythm seeded by twin primes. The resulting Hamiltonian is not split into “free particle plus interaction”. Instead, the free part is the phase/intention generator, while the remainder is the covariant relational motion over the Hilbert-Kahler/Bloch/Poincare geometry.

Epistemic status

This is a formal model-building document. The statements about $\mathbb{CP}^1$, the Bloch sphere, the Fubini–Study metric, Berry curvature, and spinorial 4π closure are standard mathematical structures. The assignments of zeta zeros to the spin poles, the extraction of tetrahedral depth from these pole labels, and the Collatz/twin-prime dynamical seed are *model axioms*. They are not claimed here as established consequences of standard quantum mechanics or cosmology. The purpose of the document is to make the model mathematically explicit enough to simulate, test, falsify, repair, or extend.

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1 Primitive data and notation

The construction uses the following primitive objects:

$$\mathcal{H} \simeq \mathbb{C}^2, \quad \text{two-level Hilbert space,} \quad (1)$$

$$\mathbb{P}(\mathcal{H}) \simeq \mathbb{CP}^1 \simeq S^2, \quad \text{projective Hilbert/Kahler manifold,} \quad (2)$$

$$(\theta, \phi) \in [0, \pi] \times [0, 2\pi), \quad \text{Bloch coordinates,} \quad (3)$$

$$\kappa := \frac{\ln 2}{24\pi}, \quad \text{information coupling constant,} \quad (4)$$

$$s = (p, p+2), \quad \text{twin-prime seed,} \quad (5)$$

$$C(n) = \begin{cases} n/2, & n \equiv 0 \pmod{2}, \\ 3n+1, & n \equiv 1 \pmod{2}, \end{cases} \quad \text{Collatz map.} \quad (6)$$

The central objective is to construct a Hamiltonian

$$\boxed{\hat{H}_s(\tau, k) = \hat{H}_{\text{phase/free},s}(\tau, k) + \hat{H}_{\perp,s}(\tau, k)} \quad (7)$$

where $\hat{H}_{\text{phase/free}}$ is generated by information/intention phase rather than by ordinary free-particle kinetic energy.

2 Hilbert space, projectivization, and the Bloch sphere

Let a normalized spinor be written as

$$|\psi(\theta, \phi)\rangle = \cos \frac{\theta}{2} |0\rangle + e^{i\phi} \sin \frac{\theta}{2} |1\rangle. \quad (8)$$

The physical ray $[\psi]$ is invariant under a global phase:

$$|\psi\rangle \sim e^{i\alpha} |\psi\rangle. \quad (9)$$

Therefore the physical state space is the projective Hilbert space

$$\mathbb{P}(\mathcal{H}) = \{|\psi\rangle : \langle\psi|\psi\rangle = 1\}/U(1) \simeq \mathbb{CP}^1 \simeq S^2. \quad (10)$$

The corresponding Bloch vector is

$$\mathbf{n} = \langle\psi|\boldsymbol{\sigma}|\psi\rangle = (\sin \theta \cos \phi, \sin \theta \sin \phi, \cos \theta), \quad (11)$$

where $\boldsymbol{\sigma} = (\sigma_x, \sigma_y, \sigma_z)$ are the Pauli matrices.

2.1 Fubini–Study metric

The Fubini–Study line element is

$$ds_{\text{FS}}^2 = \langle d\psi|d\psi\rangle\langle d\psi|d\psi\rangle - |\langle\psi|d\psi\rangle|^2. \quad (12)$$

For the spinor above one obtains

$$\boxed{ds_{\text{FS}}^2 = \frac{1}{4} (d\theta^2 + \sin^2 \theta d\phi^2)}. \quad (13)$$

Thus the projective qubit space is a round sphere of radius 1/2 in the Fubini–Study normalization.

2.2 Kahler form and Berry curvature

The Berry connection in a standard gauge may be taken as

$$\mathcal{A}_B = i \langle \psi | d\psi | \psi | d\psi \rangle = \frac{1 - \cos \theta}{2} d\phi, \quad (14)$$

up to a gauge sign convention. Its curvature is

$$F_B = d\mathcal{A}_B = \frac{1}{2} \sin \theta d\theta \wedge d\phi. \quad (15)$$

This curvature is also the Kahler form on $\mathbb{CP}^1$ in the same normalization:

$$\Omega_{\text{FS}} = F_B. \quad (16)$$

Integrating over the sphere gives

$$\int_{S^2} F_B = \int_0^{2\pi} \int_0^\pi \frac{1}{2} \sin \theta d\theta d\phi = 2\pi, \quad (17)$$

so the first Chern number of the tautological line bundle is

$$\frac{1}{2\pi} \int_{S^2} F_B = 1. \quad (18)$$

This provides the elementary topological unit of Berry phase on the Bloch sphere.

3 Euler/spin constraint and polarization

The spinorial sector is represented by the double cover

$$SU(2) \longrightarrow SO(3). \quad (19)$$

A rotation by 2π changes a spin-1/2 state by a sign, while a rotation by 4π returns it to itself:

$$\psi(\alpha + 2\pi) = -\psi(\alpha), \quad \psi(\alpha + 4\pi) = \psi(\alpha). \quad (20)$$

This is the formal origin of the 720° closure rule. In the present model this is encoded as an Euler/spin constraint:

$$\Phi_E = s \oint_\gamma \omega_E, \quad s = \frac{1}{2}, \quad (21)$$

where ω_E is an Euler/spin connection along a closed path γ .

Because $s = 1/2$ has two local signs, the state naturally carries a plus/minus polarization:

$$\mathcal{H} = \mathcal{H}_+ \oplus \mathcal{H}_-. \quad (22)$$

The poles of the Bloch sphere are then distinguished as the eigenstates of σ_z :

$$N \equiv |0\rangle, \quad S \equiv |1\rangle, \quad (23)$$

with

$$\sigma_z |0\rangle = +|0\rangle, \quad \sigma_z |1\rangle = -|1\rangle. \quad (24)$$

This gives the model its polar axis.

4 Aharonov–Bohm, Berry, Euler, and intention phase

The total geometric phase connection is taken as

$$\boxed{\mathcal{A}_{\text{ABE}} = \mathcal{A}_{\text{AB}} + \mathcal{A}_B + \mathcal{A}_E} \quad (25)$$

with

$$\mathcal{A}_{\text{AB}} = \frac{e}{\hbar} A_i(x) dx^i, \quad (26)$$

$$\mathcal{A}_B = i \langle u(q) | du(q) | u(q) | du(q) \rangle, \quad (27)$$

$$\mathcal{A}_E = \frac{1}{2} \omega_E. \quad (28)$$

The phase picked up around a loop γ is

$$\Phi_{\text{ABE}}[\gamma] = \oint_{\gamma} \mathcal{A}_{\text{ABE}}. \quad (29)$$

The intention/information phase is introduced by an operator

$$\boxed{\hat{\mathcal{I}}_s(\tau, k) = \kappa \hat{W}_s + \delta \hat{\mathcal{I}}_0(\tau, k), \quad \kappa = \frac{\ln 2}{24\pi}.} \quad (30)$$

Here $\hat{W}_s$ is the structural operator carrying winding, seed, and information number in the sector $s = (p, p+2)$, while $\delta \hat{\mathcal{I}}_0$ is the zero-level fluctuation term.

The accumulated intention phase over Collatz steps is

$$\Theta_{\mathcal{I}}(K) = \sum_{k=0}^{K-1} \rho_s(k) \langle \hat{\mathcal{I}}_s(\tau_k, k) \rangle, \quad (31)$$

where $\rho_s(k)$ is the Collatz rhythm extracted from the twin-prime seed.

5 Cosmological Lambda and neutrino epsilon closure

The Euler–Berry closure is not left as an abstract degree plus arbitrary defect. In this formulation it is fixed by two physical-symbolic sectors:

$$D \equiv D_{\Lambda}, \quad \epsilon \equiv \epsilon_{\nu}. \quad (32)$$

Since the cosmological constant has dimension L^{-2} , it enters the phase equation through a dimensionless curvature area:

$$\boxed{D_{\Lambda}[\Sigma] = \frac{1}{2\pi} \int_{\Sigma} \Lambda dA_{\text{phys}}.} \quad (33)$$

For constant Λ and physical area A_{Σ} this reduces to

$$\boxed{D_{\Lambda} = \frac{\Lambda A_{\Sigma}}{2\pi}.} \quad (34)$$

The neutrino phase defect is modeled by the standard oscillatory phase between mass sectors i, j :

$$\Delta\phi_{ij}^{(\nu)} = \frac{\Delta m_{ij}^2 L}{2E} \quad (\hbar = c = 1), \quad (35)$$

so that

$$\boxed{\epsilon_{\nu}^{ij} = \frac{1}{2\pi} \Delta\phi_{ij}^{(\nu)} = \frac{\Delta m_{ij}^2 L}{4\pi E}.} \quad (36)$$

For multiple channels one may use

$$\epsilon_\nu = \sum_{i < j} w_{ij} \frac{\Delta m_{ij}^2 L}{4\pi E}, \quad \sum_{i < j} w_{ij} = 1. \quad (37)$$

The closure constraint is then

$$\boxed{\Phi_{AB} + \Phi_B + \Phi_E + \Theta_{\mathcal{I}} = 2\pi (m + D_\Lambda + \epsilon_\nu), \quad m \in \mathbb{Z}.} \quad (38)$$

Equivalently, selecting one branch m ,

$$\boxed{\Delta_{\text{closure}} := \frac{\Phi_{AB} + \Phi_B + \Phi_E + \Theta_{\mathcal{I}}}{2\pi} - m - D_\Lambda - \epsilon_\nu = 0.} \quad (39)$$

In computations one monitors Δ_{closure} as the Euler–Berry–Lambda–neutrino phase defect.

6 Zeta-zero polar axiom

The following step is not standard geometry; it is a model axiom.

Axiom 1 (Zeta-polar labeling). *The north and south spin poles of the Bloch sphere are labeled by conjugate nontrivial zeta-zero data:*

$$N \mapsto \frac{1}{2} + i\gamma_n, \quad S \mapsto \frac{1}{2} - i\gamma_n, \quad (40)$$

where $\zeta(1/2 + i\gamma_n) = 0$.

The pole labels define a spectral boundary condition for the phase bundle:

$$\Psi(N) = e^{i\alpha_n} \Psi_N, \quad \Psi(S) = e^{-i\alpha_n} \Psi_S, \quad (41)$$

where one may set, for example,

$$\alpha_n = \gamma_n \pmod{2\pi}. \quad (42)$$

This converts zeta-zero data into a polar phase label. It does not assert that standard zeta zeros are literally located on the physical Bloch sphere; it defines how the model places spectral data at the poles.

7 Tetrahedral depth from polar spectral data

A tetrahedral frame inside the Bloch sphere is given by four unit vectors

$$\mathbf{v}_1 = \frac{(1, 1, 1)}{\sqrt{3}}, \quad \mathbf{v}_2 = \frac{(1, -1, -1)}{\sqrt{3}}, \quad \mathbf{v}_3 = \frac{(-1, 1, -1)}{\sqrt{3}}, \quad \mathbf{v}_4 = \frac{(-1, -1, 1)}{\sqrt{3}}, \quad (43)$$

which satisfy

$$\boxed{\mathbf{v}_i \cdot \mathbf{v}_j = \begin{cases} 1, & i = j, \\ -1/3, & i \neq j. \end{cases}} \quad (44)$$

The tetrahedral coordinates of a Bloch vector $\mathbf{n}$ are defined by the qubit SIC map

$$\boxed{p_j(\mathbf{n}) = \frac{1}{4} (1 + \mathbf{n} \cdot \mathbf{v}_j), \quad \sum_{j=1}^4 p_j = 1.} \quad (45)$$

This assigns every Bloch-sphere state to a tetrahedral probability coordinate. The model-specific zeta-pole data may then be used to label the tetrahedral vertices:

$$\mathbf{v}_j \mapsto \frac{1}{2} + i\gamma_j, \quad j = 1, 2, 3, 4, \quad (46)$$

or, in a sign-symmetric version, to conjugate pairs attached to opposite tetrahedral sectors.

The “depth” of the sphere is represented by moving from the surface S^2 into the Bloch ball

$$B^3 = \{\mathbf{r} \in \mathbb{R}^3 : \|\mathbf{r}\| \leq 1\}, \quad (47)$$

with tetrahedral barycentric coordinates

$$\mathbf{r} = \sum_{j=1}^4 b_j \mathbf{v}_j, \quad \sum_{j=1}^4 b_j = 1. \quad (48)$$

Interference between tetrahedral sectors is encoded by the Gram matrix

$$G_{ij} = \frac{1 + \mathbf{v}_i \cdot \mathbf{v}_j}{2}, \quad (49)$$

so that

$$G_{ii} = 1, \quad G_{ij} = \frac{1}{3} \quad (i \neq j). \quad (50)$$

This gives a concrete tetrahedral internal coordinate system for the phase bundle.

Zeta-labelled poles and tetrahedral depth frame

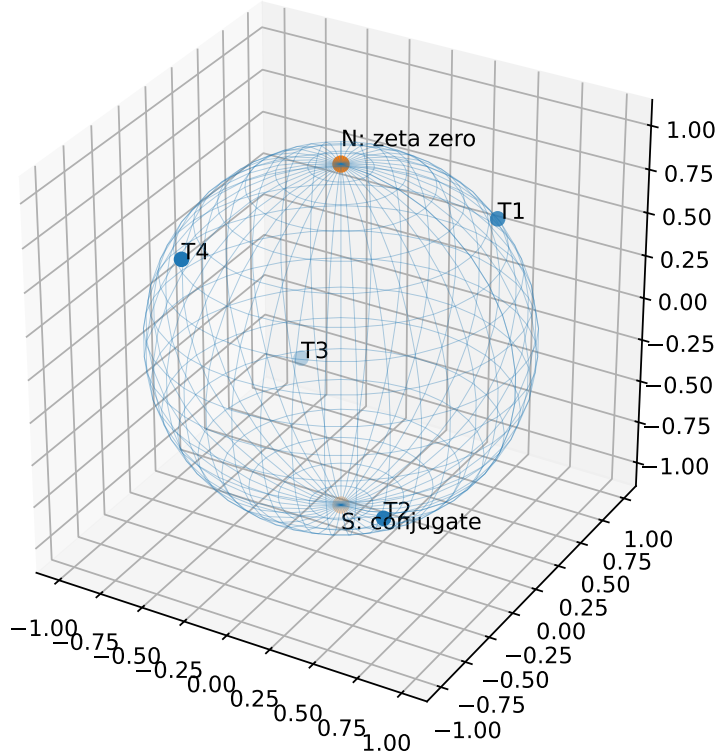


Figure 1: Model axiom: zeta-zero labels placed at the spin poles and a tetrahedral depth frame placed inside the Bloch sphere.

8 Poincare disk as the tetrahedral interference surface

Let $\Pi_T \subset B^3$ be a two-dimensional plane selected by two independent tetrahedral interference modes. Choose orthonormal coordinates (u, v) on this plane and define

$$R^2 = u^2 + v^2 < 1. \quad (51)$$

The associated Poincare disk is

$$\mathbb{D} = \{z = u + iv : |z| < 1\} \quad (52)$$

with metric

$$ds_{\mathbb{D}}^2 = \frac{4|dz|^2}{(1 - |z|^2)^2}. \quad (53)$$

The disk is used as the local dynamical arena for Collatz motion induced by the twin-prime seed.

9 Twin-prime Collatz rhythm and disk dynamics

Let the free seed be a twin-prime pair

$$s = (p, p + 2), \quad p, p + 2 \in \mathbb{P}. \quad (54)$$

Define the paired Collatz orbit

$$\mathcal{O}_s(k) = \left(C^k(p), C^k(p + 2) \right) =: (a_k, b_k). \quad (55)$$

A bounded positive rhythm may be extracted from the pair by any fixed rule. One convenient reference rule is

$$\rho_s(k) = \frac{\log(1 + a_k) + \log(1 + b_k)}{1 + \log(1 + a_k + b_k)}. \quad (56)$$

The Poincare-disk step is modeled by an automorphism

$$z_{k+1} = e^{i\alpha_k} \frac{z_k + \beta_k}{1 + \overline{\beta_k} z_k}, \quad |\beta_k| < 1, \quad (57)$$

where

$$\alpha_k = \alpha(\rho_s(k), \kappa, k), \quad \beta_k = \beta(\rho_s(k), \kappa, k). \quad (58)$$

The twin-prime seed therefore determines the rhythm, the rhythm determines the disk automorphism, and the disk orbit modulates the phase generator.

10 Relational Lagrangian with phase as the primary free sector

Let q^a be coordinates on the relational/Kahler base. Let $\chi \in U(1)$ be the phase coordinate. Define the covariant phase velocity

$$D_t \chi = \dot{\chi} + \mathcal{A}_a(q) \dot{q}^a, \quad (59)$$

where

$$\mathcal{A} = \mathcal{A}_{\text{ABE}} + \mathcal{A}_{\mathcal{I}}. \quad (60)$$

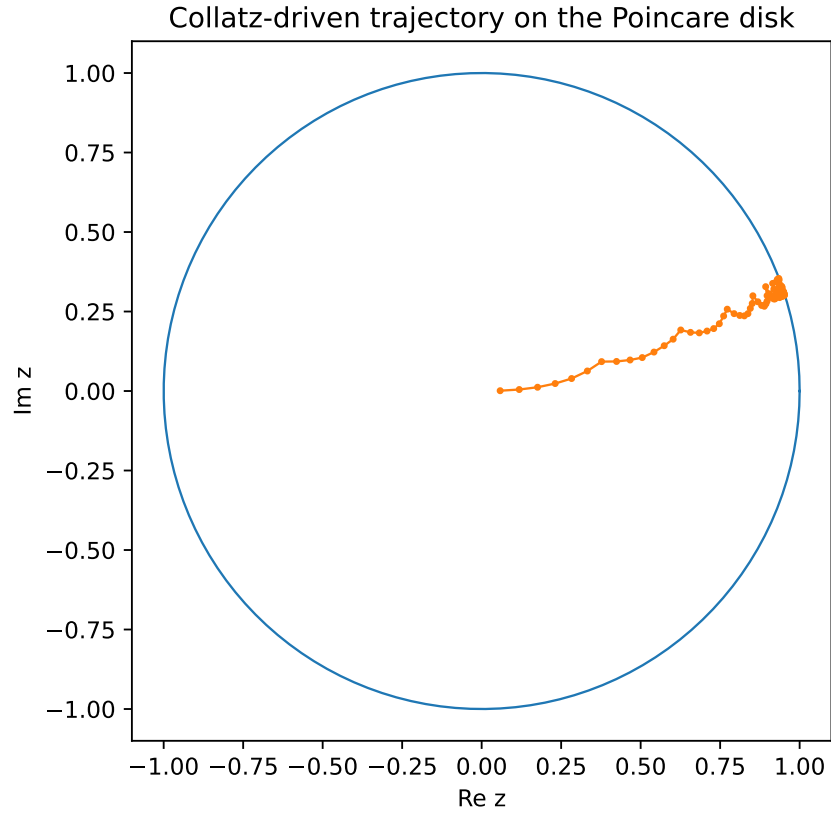


Figure 2: Reference simulation: Collatz-driven trajectory on the Poincare disk for seed (11, 13).

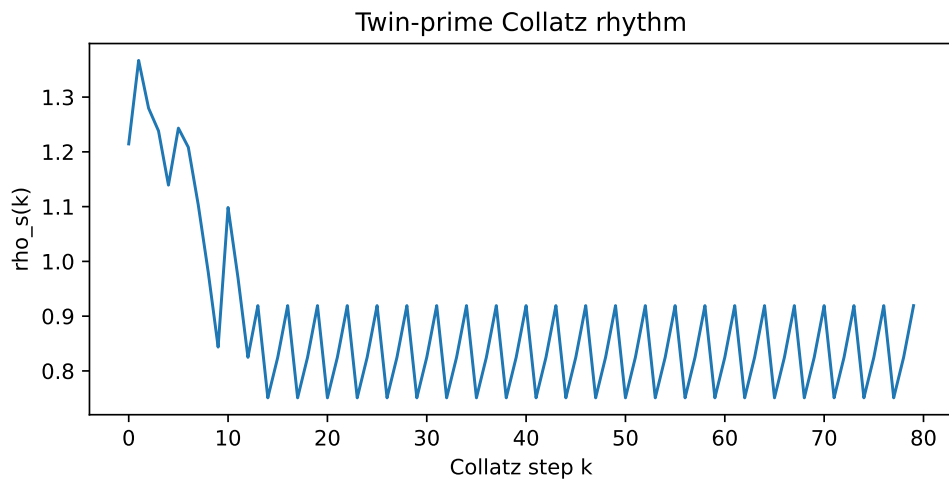


Figure 3: Reference rhythm $\rho_s(k)$ extracted from the twin-prime Collatz pair.

A minimal relational Lagrangian is

$$L = \frac{1}{2}g_{ab}(q)\dot{q}^a\dot{q}^b + \frac{I_\phi}{2}(D_t\chi)^2 + J_0D_t\chi - V(q). \quad (61)$$

The phase momentum is

$$J = \frac{\partial L}{\partial \dot{\chi}} = I_\phi D_t\chi + J_0, \quad (62)$$

so

$$D_t\chi = \frac{J - J_0}{I_\phi}. \quad (63)$$

The relational momentum is

$$p_a = \frac{\partial L}{\partial \dot{q}^a} = g_{ab}\dot{q}^b + J\mathcal{A}_a. \quad (64)$$

Thus the covariant momentum is

$$\Pi_a = p_a - J\mathcal{A}_a. \quad (65)$$

Solving for $\dot{q}^a$ gives

$$\dot{q}^a = g^{ab}\Pi_b. \quad (66)$$

The Hamiltonian is

$$H = p_a\dot{q}^a + J\dot{\chi} - L \quad (67)$$

$$= \frac{(J - J_0)^2}{2I_\phi} + \frac{1}{2}g^{ab}(p_a - J\mathcal{A}_a)(p_b - J\mathcal{A}_b) + V(q). \quad (68)$$

Therefore

$$H = H_\Phi + H_\perp, \quad (69)$$

where

$$H_\Phi = \frac{(J - J_0)^2}{2I_\phi}, \quad H_\perp = \frac{1}{2}g^{ab}\Pi_a\Pi_b + V(q). \quad (70)$$

This is the classical phase-first split.

11 Replacement of the bare phase rotor by the intention/information generator

The bare rotor J is now replaced by the intention/information generator. The quantized phase generator is not merely $\kappa\mathbf{1}$; it is

$$\hat{\mathcal{I}}_s(\tau, k) = \kappa\hat{W}_s + \delta\hat{\mathcal{I}}_0(\tau, k). \quad (71)$$

The legal domain is

$$D(\hat{\mathcal{I}}_s) = \left\{ \Psi_s \in \mathcal{H}_s : \hat{W}_s\Psi_s \in \mathcal{H}_s, \Psi_s \text{ satisfies ABE/Euler-Berry-Lambda-neutrino closure} \right\}. \quad (72)$$

The free phase Hamiltonian is then

$$\hat{H}_{\text{phase/free},s}(\tau, k) = \frac{\hbar}{\Delta\tau_k}\rho_s(k) \left[\kappa\hat{W}_s + \delta\hat{\mathcal{I}}_0(\tau, k) \right]. \quad (73)$$

This is the precise sense in which the free part is phase/intention rather than free-particle kinetic energy.

12 Final Hamiltonian on the Hilbert-Kahler/Bloch/Poincare structure

Let the full covariant derivative include the Aharonov–Bohm, Berry, Euler, and intention connections:

$$\hat{\Pi}_A = -i\hbar\nabla_A - \hbar\mathcal{A}_A^{\text{ABE}} - \lambda_s(k)\mathcal{I}_A, \quad (74)$$

where A ranges over the chosen coordinates of the Kahler/Bloch/Poincare base. With the block metric

$$G = g_{\text{FS}} \oplus g_{\mathbb{D}} \oplus g_{\text{rel}}, \quad (75)$$

we obtain

$$\hat{H}_s(\tau, k) = \frac{\hbar}{\Delta\tau_k} \rho_s(k) \left[\frac{\ln 2}{24\pi} \hat{W}_s + \delta\hat{\mathcal{I}}_0(\tau, k) \right] + \frac{1}{2} G^{AB} \hat{\Pi}_A \hat{\Pi}_B + V_{\zeta, T, \Lambda, \nu}(q, z, \tau, k). \quad (76)$$

The potential may be decomposed as

$$V_{\zeta, T, \Lambda, \nu} = V_T + V_{\zeta} + V_{\Lambda} + V_{\nu} + V_{\text{guard}}, \quad (77)$$

where V_T is tetrahedral-depth structure, V_{ζ} is polar spectral labeling, V_{Λ} is the cosmological curvature contribution, V_{ν} is the neutrino phase-defect contribution, and V_{guard} is any additional constraint/regularization term.

The admissible states are constrained by

$$\frac{\Phi_{\text{AB}} + \Phi_B + \Phi_E + \Theta_{\mathcal{I}}}{2\pi} - m - D_{\Lambda} - \epsilon_{\nu} = 0. \quad (78)$$

If the constraint is not exactly satisfied, one monitors the defect

$$\varepsilon_{\text{clos}} = \frac{\Phi_{\text{AB}} + \Phi_B + \Phi_E + \Theta_{\mathcal{I}}}{2\pi} - m - D_{\Lambda} - \epsilon_{\nu}. \quad (79)$$

The branch m is chosen by nearest integer closure:

$$m = \text{round} \left(\frac{\Phi_{\text{AB}} + \Phi_B + \Phi_E + \Theta_{\mathcal{I}}}{2\pi} - D_{\Lambda} - \epsilon_{\nu} \right). \quad (80)$$

13 Numerical reference checks

The included workbench contains three small scripts.

1. `geometry_checks.py`: verifies the Berry curvature integral on $\mathbb{CP}^1$ numerically and records the spinorial $2\pi/4\pi$ closure convention.
2. `zeta_tetrahedral_map.py`: constructs the tetrahedral SIC frame, labels the poles by zeta-zero data as a model axiom, and exports the tetrahedral geometry.
3. `simulate_collatz_poincare.py`: generates a twin-prime Collatz rhythm, drives a Poincare-disk automorphism, accumulates intention phase, and tracks the Euler–Berry–Lambda–neutrino closure defect.

The Berry curvature check gives

$$\frac{1}{2\pi} \int_{S^2} F_B \approx 1, \quad (81)$$

up to numerical quadrature error. This validates the normalization used by the document.

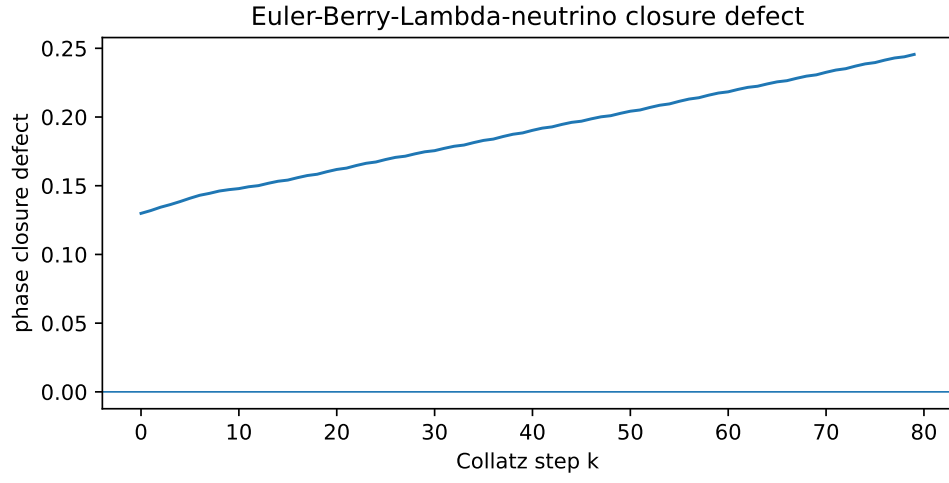


Figure 4: Reference simulation of the closure defect $\varepsilon_{\text{clos}}$ under Collatz rhythm, intention phase, D_Λ , and ϵ_ν .

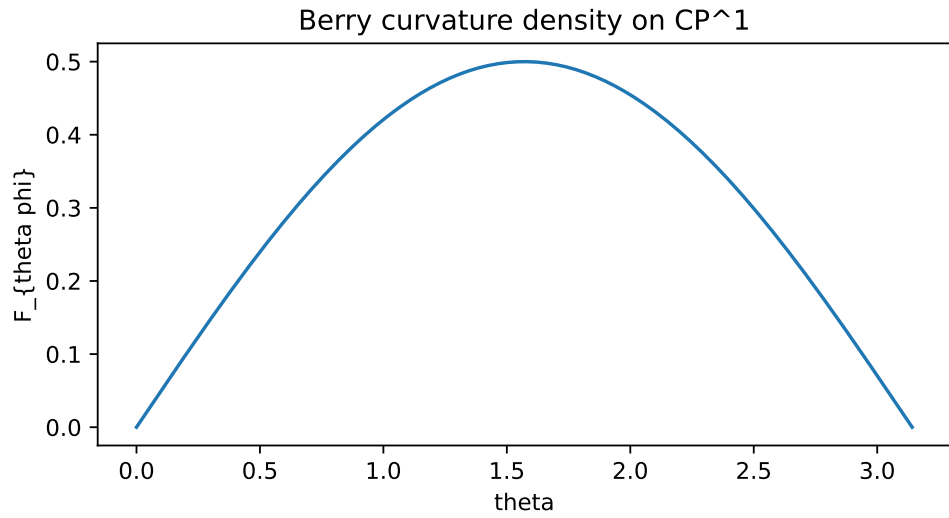


Figure 5: Berry curvature density $F_{\theta\phi} = \frac{1}{2} \sin \theta$ on $\mathbb{CP}^1$.

14 Compact derivation chain

The complete chain can be written as

$$\boxed{\mathcal{H} \rightarrow \mathbb{P}(\mathcal{H}) \simeq \mathbb{CP}^1 \rightarrow (S^2, g_{\text{FS}}, F_B) \rightarrow s = \frac{1}{2} \rightarrow (N, S) \rightarrow \zeta\text{-polar labels} \rightarrow T^3 \text{ depth} \rightarrow \mathbb{D} \rightarrow \mathcal{O}_s(k) \rightarrow \rho_s(k) \rightarrow \hat{H}_s.} \quad (82)$$

In words:

1. Projectivizing the Hilbert space gives a Kahler manifold $\mathbb{CP}^1$.
2. $\mathbb{CP}^1$ is the Bloch sphere with Fubini–Study metric and Berry curvature.
3. The Euler/spin constraint selects spin 1/2 and 4π closure.
4. Spin polarization distinguishes north and south poles.
5. The model labels those poles by conjugate zeta-zero data.
6. The pole-labeled sphere is given tetrahedral depth through a SIC tetrahedral frame.
7. Tetrahedral interference selects a Poincare disk.
8. A twin-prime seed generates a paired Collatz orbit.
9. The Collatz orbit defines the rhythm $\rho_s(k)$.
10. The rhythm drives the intention/information phase generator.
11. The resulting Hamiltonian has phase/intention as its free part and covariant relational motion as the remainder.

15 Open definitions that must become canonical

The model is now explicit, but several components must still be fixed canonically before the framework can be treated as a stable formal system:

1. The exact operator $\hat{W}_s$: winding number, information number, Collatz orbital operator, or a composite of these.
2. The exact rhythm map $\rho_s(k)$: the reference logarithmic map is only a stable default.
3. The zero-level fluctuation law $\delta\hat{\mathcal{I}}_0$: deterministic, stochastic, Kuramoto-like, HTRI-like, or spectral.
4. The zeta-polar assignment: which zeros, which branch convention, and how conjugate pole labels are transported to tetrahedral vertices.
5. The plane Π_T defining the Poincare disk from tetrahedral interference.
6. The precise physical scale converting Λ into D_Λ through A_{phys} .
7. The neutrino channel weights w_{ij} defining ϵ_ν .

A Code: CP1 Berry geometry check

```
#!/usr/bin/env python3
"""Numerical checks for CP1 / Bloch sphere Berry geometry."""
from __future__ import annotations
import json
import math
from pathlib import Path
import numpy as np
import matplotlib.pyplot as plt

OUT = Path(__file__).resolve().parents[1] / "results"
FIG = Path(__file__).resolve().parents[1] / "figures"
OUT.mkdir(exist_ok=True)
FIG.mkdir(exist_ok=True)

def berry_curvature(theta):
    # For spin 1/2 in the north gauge: F = 1/2 sin(theta) dtheta wedge dphi.
    return 0.5 * np.sin(theta)

def main(n_theta=600, n_phi=600):
    theta = np.linspace(0, math.pi, n_theta)
    phi = np.linspace(0, 2 * math.pi, n_phi)
    dtheta = theta[1] - theta[0]
    dphi = phi[1] - phi[0]
    integral = float(np.sum(berry_curvature(theta)[: , None]) * dtheta * dphi)
    chern = integral / (2 * math.pi)
    result = {
        "berry_curvature_integral_numeric": integral,
        "expected_integral": 2 * math.pi,
        "chern_number_numeric": chern,
        "spinor_closure": {"psi_2pi": "-psi", "psi_4pi": "+psi"},
    }
    (OUT / "cp1_berry_geometry_check.json").write_text(json.dumps(result, indent=2), encoding="utf-8")

    tt = np.linspace(0, math.pi, 400)
    plt.figure(figsize=(6.0, 3.4))
    plt.plot(tt, berry_curvature(tt), linewidth=1.5)
    plt.xlabel("theta")
    plt.ylabel("F_{theta phi}")
    plt.title("Berry curvature density on CP1")
    plt.tight_layout()
    plt.savefig(FIG / "berry_curvature_density.pdf")
    plt.close()

if __name__ == "__main__":
    main()
```

B Code: zeta-polar tetrahedral map

```
#!/usr/bin/env python3
"""Tetrahedral depth map with zeta-zero pole labels as a model axiom."""
from __future__ import annotations
import json
import math
from pathlib import Path
import numpy as np
import matplotlib.pyplot as plt

OUT = Path(__file__).resolve().parents[1] / "results"
FIG = Path(__file__).resolve().parents[1] / "figures"
OUT.mkdir(exist_ok=True)
FIG.mkdir(exist_ok=True)

# First four imaginary parts of nontrivial zeta zeros, used as spectral labels.
# These are labels in this model, not a derivation from standard CP^1 geometry.
ZETA_GAMMA = [14.134725141, 21.022039639, 25.010857580, 30.424876126]

def tetra_vertices():
    verts = np.array([
        [1, 1, 1],
        [1, -1, -1],
        [-1, 1, -1],
        [-1, -1, 1],
    ], dtype=float)
    return verts / np.linalg.norm(verts, axis=1)[:, None]

def sic_probs(n, verts):
    # Tetrahedral qubit SIC coordinates. Sum_j p_j = 1.
    return np.array([(1.0 + float(np.dot(n, v))) / 4.0 for v in verts])

def main():
    verts = tetra_vertices()
    gram = [[float(np.dot(verts[i], verts[j])) for j in range(4)] for i in range(4)]
    north = np.array([0.0, 0.0, 1.0])
    south = np.array([0.0, 0.0, -1.0])
    data = {
        "zeta_pole_axiom": {
            "north_pole_label": "1/2 + i gamma_1",
            "south_pole_label": "1/2 - i gamma_1",
            "gamma_1": ZETA_GAMMA[0],
        },
        "tetrahedral_vertices": verts.tolist(),
        "dot_product_matrix": gram,
        "sic_probabilities_north": sic_probs(north, verts).tolist(),
        "sic_probabilities_south": sic_probs(south, verts).tolist(),
        "zeta_labels_for_vertices": [f"1/2 + i*{g}" for g in ZETA_GAMMA],
    }
    (OUT / "zeta_tetrahedral_map.json").write_text(json.dumps(data, indent=2), encoding="utf-8")

    # Spherical projection figure.
    fig = plt.figure(figsize=(5.2, 5.2))
    ax = fig.add_subplot(111, projection="3d")
    u = np.linspace(0, 2 * np.pi, 40)
    v = np.linspace(0, np.pi, 20)
    x = np.outer(np.cos(u), np.sin(v))
    y = np.outer(np.sin(u), np.sin(v))
    z = np.outer(np.ones_like(u), np.cos(v))
    ax.plot_wireframe(x, y, z, linewidth=0.25, alpha=0.35)
    ax.scatter(verts[:,0], verts[:,1], verts[:,2], s=35)
    for i, p in enumerate(verts):
        ax.text(p[0], p[1], p[2], f"T{i+1}")
    ax.scatter([0,0],[0,0],[1,-1], s=45)
    ax.text(0,0,1.1,"N: zeta zero")
    ax.text(0,0,-1.15,"S: conjugate")
    ax.set_title("Zeta-labelled poles and tetrahedral depth frame")
    ax.set_box_aspect((1,1,1))
```

```
plt.tight_layout()
plt.savefig(FIG / "zeta_tetrahedral_frame.pdf")
plt.close()

if __name__ == "__main__":
    main()
```


C Code: Collatz-Poincare phase workbench

```
#!/usr/bin/env python3
"""
Collatz-Poincare phase workbench for the phase-intention Hamiltonian.
This is a structural simulation, not a physical proof.
"""
from __future__ import annotations
import json
import math
from pathlib import Path
import numpy as np
import matplotlib.pyplot as plt

OUT = Path(__file__).resolve().parents[1] / "results"
FIG = Path(__file__).resolve().parents[1] / "figures"
OUT.mkdir(exist_ok=True)
FIG.mkdir(exist_ok=True)

KAPPA = math.log(2.0) / (24.0 * math.pi)

def collatz(n: int) -> int:
    return n // 2 if n % 2 == 0 else 3 * n + 1

def twin_collatz(seed=(11, 13), steps=80):
    a, b = seed
    seq = []
    for k in range(steps):
        seq.append((k, a, b))
        a, b = collatz(a), collatz(b)
    return seq

def rhythm(a: int, b: int) -> float:
    # Bounded positive rhythm extracted from the twin Collatz pair.
    # The log prevents large odd jumps from dominating the phase.
    return (math.log1p(a) + math.log1p(b)) / (1.0 + math.log1p(a + b))

def disk_step(z: complex, rho: float, k: int) -> complex:
    # A Poincare disk automorphism  $z \rightarrow \exp(i \alpha) (z + \beta) / (1 + \text{conj}(\beta)z)$ .
    alpha = (KAPPA * rho * (1 + (k % 7))) % (2 * math.pi)
    beta_abs = min(0.35, 0.04 + 0.015 * rho)
    beta = beta_abs * complex(math.cos(alpha / 2), math.sin(alpha / 2))
    return complex(math.cos(alpha), math.sin(alpha)) * (z + beta) / (1 + beta.conjugate() * z)

def main(seed=(11, 13), steps=80, phi_abe=1.75, d_lambda=0.12, eps_nu=0.031):
    seq = twin_collatz(seed, steps)
    z = 0.0 + 0.0j
    rows = []
    theta_i = 0.0
    for k, a, b in seq:
        rho = rhythm(a, b)
        # A simple winding observable  $W_s(k)$ ; replace with canonical  $W_s$  in later work.
        w = ((a - b) % 12) / 12.0 + 0.5
        delta_i0 = 0.07 * KAPPA * math.sin(2 * math.pi * k / 17.0)
        theta_i += rho * (KAPPA * w + delta_i0)
        z = disk_step(z, rho, k)
        phi_total = phi_abe + theta_i
        closure = phi_total - 2 * math.pi * (d_lambda + eps_nu)
        branch = round(closure / (2 * math.pi))
        defect = closure / (2 * math.pi) - branch
        rows.append({
            "k": k, "a": a, "b": b, "rho": rho, "W": w,
            "theta_i": theta_i, "disk_re": z.real, "disk_im": z.imag,
            "phi_total": phi_total, "closure_branch": branch, "closure_defect": defect,
        })

    (OUT / "collatz_poincare_phase.json").write_text(json.dumps({
```

```

        "seed": seed,
        "kappa": KAPPA,
        "phi_ABE": phi_abe,
        "D_Lambda": d_lambda,
        "epsilon_nu": eps_nu,
        "rows": rows,
    }, indent=2), encoding="utf-8")

# Figure 1: disk trajectory.
xs = [r["disk_re"] for r in rows]
ys = [r["disk_im"] for r in rows]
t = np.linspace(0, 2 * np.pi, 400)
plt.figure(figsize=(5.2, 5.2))
plt.plot(np.cos(t), np.sin(t), linewidth=1)
plt.plot(xs, ys, marker="o", markersize=2, linewidth=1)
plt.gca().set_aspect("equal", adjustable="box")
plt.xlabel("Re z")
plt.ylabel("Im z")
plt.title("Collatz-driven trajectory on the Poincare disk")
plt.tight_layout()
plt.savefig(FIG / "poincare_collatz_trajectory.pdf")
plt.close()

# Figure 2: closure defect.
plt.figure(figsize=(6.5, 3.4))
plt.plot([r["k"] for r in rows], [r["closure_defect"] for r in rows], linewidth=1.4)
plt.axhline(0.0, linewidth=0.8)
plt.xlabel("Collatz step k")
plt.ylabel("phase closure defect")
plt.title("Euler-Berry-Lambda-neutrino closure defect")
plt.tight_layout()
plt.savefig(FIG / "closure_defect.pdf")
plt.close()

# Figure 3: rhythm.
plt.figure(figsize=(6.5, 3.4))
plt.plot([r["k"] for r in rows], [r["rho"] for r in rows], linewidth=1.4)
plt.xlabel("Collatz step k")
plt.ylabel("rho_s(k)")
plt.title("Twin-prime Collatz rhythm")
plt.tight_layout()
plt.savefig(FIG / "collatz_rhythm.pdf")
plt.close()

if __name__ == "__main__":
    main()

```